Zero-shot text Restoration via Masked Diffusion Models: Ablation Study on Text Infilling

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Abstract

Autoregressive Large Language Models drive most recent NLP progress, but their strictly left-to-right generation struggles with hard-constrained Controlled Text Generation tasks such as zero-shot text infilling and restoration, where standard fixes (prompt engineering, task-specific fine-tuning) still fail to guarantee exact adherence to bidirectional context and token boundaries. We explore non-autoregressive masked diffusion models, which reformulate generation as a discrete denoising process and natively enforce such constraints via absorbing states, with no additional training. Using synthetically corrupted subsets of WikiText-103, we systematically compare masking-based diffusion against autoregressive Fill-in-the-Middle prompting, ablating masking ratios, corruption topology (random vs. span masking), and inference efficiency (denoising timesteps). We evaluate semantic coherence (BERTScore), stylistic overlap (ROUGE-L), fluency (Perplexity), and computational trade-offs, aiming to empirically validate whether discrete diffusion better preserves bidirectional context in hard-constraint text generation.

1 Task description/Problem statement

The task addressed in this work is **zero-shot text restoration**, also referred to as **text infilling**: given a document in which one or more contiguous or scattered spans of text have been removed and replaced with a placeholder (a *mask*), the goal is to reconstruct content for each masked span such that the resulting document is fluent, coherent with the surrounding bidirectional context, and semantically close to the original (uncorrupted) text. Unlike free-form continuation, this is a **hard-constrained generation** task: the model must produce output that respects fixed left and right context, and, in the span-masking case, must also implicitly determine how much content to generate to plausibly fill the gap.

Formally, given an original document $x = (w_1, \dots, w_n)$ composed of n words, a *corruption function* selects a subset of word positions $M \subset \{1, \dots, n\}$ (either scattered, for random token masking, or a single contiguous run, for span masking) and replaces them with a mask token, producing a corrupted document $\tilde{x}$. A restoration model f must then predict $\hat{x}_M = f(\tilde{x})$, an estimate of the original content at the masked positions, so that the reconstructed document $\hat{x}$ is evaluated against x.

Standard autoregressive (AR) language models generate text strictly left-to-right, conditioning each token only on previously generated tokens. This makes them poorly suited to infilling: injecting a right-hand context (the suffix after the mask) requires ad hoc reformulations such as Fill-in-the-Middle (FIM) prompting, in which the prefix and suffix are concatenated with special separator tokens and the model is trained (or prompted) to predict the missing middle segment [2]. While workable, FIM does not guarantee that the generated fill will match the length or exact boundary of the removed span, and conditioning on the suffix is only approximate, since the underlying model still generates autoregressively rather than jointly reasoning over both directions of context.

Masked diffusion language models (MDLMs) offer a native alternative for this task [1, 8]. These models reformulate text generation as a discrete denoising process: starting from a sequence in which some positions are in an *absorb-*

*Code available at: https://github.com/TommasoFederici/Zero-Shot-text-Restoration-via-Masked-Diffusion-Model *ing* (masked) state, the model iteratively predicts a joint, bidirectional distribution over all masked positions simultaneously, progressively resolving them over a number of denoising timesteps [8, 5]. Because masking is already the model's native corruption/generative mechanism, hard constraints on which positions are fixed (unmasked context) versus to-be-generated (masked spans) are enforced exactly, at every step of the reverse process, without any task-specific fine-tuning or prompt engineering. This project investigates whether this architectural property translates into a measurable empirical advantage over AR FIM prompting for zero-shot text restoration.

1.1 Examples

Consider the sentence:

> *"The quick brown fox jumps over the lazy dog near the river."*

Under **span masking** at a 30% corruption ratio, a contiguous run of words is replaced by a single mask placeholder:

> Input: *"The quick brown fox jumps* `<mask>` *near the river."*
> Expected output (reconstructed): *"The quick brown fox jumps over the lazy dog near the river."*

Under **random token masking** at the same ratio, individual words are masked independently and non-contiguously:

> Input: *"The* `<mask>` *brown fox* `<mask>` *over* `<mask>` *lazy dog* `<mask>` *the river."*
> Expected output: the original words *"quick"*, *"jumps"*, *"the" "near"* restored at their respective positions.

In both cases the restoration model only observes the corrupted text and must infer the missing content solely from the surrounding bidirectional context, with no access to the ground truth.

1.2 Real-world applications

Zero-shot text restoration underpins several practical applications: recovering corrupted text in digitized historical documents and OCR pipelines, and context-aware code/document editing tools that fill a selected region (as in FIM-based code completion [2]). More broadly, respecting hard positional and length constraints while remaining faithful to bidirectional context is central to controlled text generation tasks such as data augmentation for low-resource settings.

2 Related work

Existing approaches to text infilling are dominated by autoregressive (AR) language models adapted via the Fill-in-the-Middle (FIM) objective [2], which underlies strong code-oriented baselines such as Qwen2.5-Coder [9]; however, FIM's left-to-right decoding does not natively enforce exact-length or bidirectional constraints. Discrete diffusion models offer an alternative paradigm: Austin et al. [1] introduced structured denoising diffusion over discrete state-spaces (D3PM), later refined into continuous-time score-based formulations such as SEDD [5] and absorbing-state masked diffusion such as MDLM [8], which we adopt as our diffusion baseline for its architectural simplicity and competitive perplexity. Unlike AR-FIM, masked diffusion enforces bidirectional context and exact token boundaries natively at every denoising step, motivating our zero-shot comparison between the two paradigms.

3 Datasets and benchmarks

Text infilling lacks a standardized benchmark; the literature instead relies on synthetically corrupted subsets of large text corpora, since ground-truth continuations are trivially available by masking naturally occurring text [2, 8]. We build our benchmark from WikiText-103 [6], a corpus of verified Wikipedia articles free of the preprocessing artifacts present in earlier corpora such as the Penn Treebank. We apply two corruption topologies: *span masking*, removing a contiguous run of tokens (mirroring FIM's prefix-suffix-middle setup [2]), and *random token masking*, independently masking scattered positions (mirroring BERT-style pretraining and MDLM's native absorbing-state corruption [3, 8]). Varying the masking ratio (10%, 30%, 50%) over both topologies yields a controlled ablation grid isolating corruption severity and structure.

4 Existing tools, libraries, papers with code

Beyond the underlying architectures, each model is released with reference implementations and checkpoints that we build directly on. Sahoo et al.

release the official MDLM codebase together with the `kuleshov-group/mdlm-owt` checkpoint via HuggingFace, which we use as our zero-shot diffusion restorer without any fine-tuning [8]. Qwen2.5-Coder is distributed at multiple scales through the HuggingFace `transformers` library [9]; we use its smallest checkpoint as our FIM baseline. We further rely on HuggingFace `datasets` for loading WikiText-103, on the reference implementations of ROUGE [4] and BERTScore [10] for evaluation, and on a HuggingFace-hosted GPT-2 checkpoint [7] for the perplexity metric.

5 State-of-the-art evaluation

The two paradigms compared here have historically been evaluated under different, non-overlapping protocols. MDLM and SEDD are primarily evaluated via zero-shot perplexity under an external language model (e.g., GPT-2) on held-out corpora; SEDD additionally shows qualitative infilling examples, but neither evaluates infilling on natural-language benchmarks [8, 5]. AR-FIM models, in contrast, are evaluated on code-infilling benchmarks such as HumanEval-Infilling via pass@k against unit tests [2] – a setting easier than natural-language infilling, since correctness is checked programmatically rather than via surface similarity to a reference. Neither line of work therefore offers a protocol for scoring restoration quality on generic natural-language text under a shared masking scheme, motivating the metric choices detailed in Section 6.

6 Comparative evaluation

Dataset. We sample 100 documents from the training split of WikiText-103 [6] (fixed seed 42), filtered to 10–500 words to keep per-example latency comparable across the grid while still exercising multi-sentence context. Each document is corrupted twice, once per corruption topology from Section 3, at three masking ratios (10%, 30%, 50%), yielding $100 \times 2 \times 3 = 600$ (corrupted, original) pairs per restoration system.

Systems and baselines. We compare two zero-shot restoration systems, neither fine-tuned on WikiText-103 or on the restoration task. **MDLM** uses the public `kuleshov-group/mdlm-owt` checkpoint [8] ($\sim$125M parameters), denoised for a variable number of timesteps (4, 8, 16), used as a third ablation axis to trade inference cost for quality; all masked positions in a document are resolved jointly via the model's own absorbing-state reverse-diffusion update, ported from the official `kuleshov-group/mdlm` repository's `diffusion.py` (since the released checkpoint documents only unconditional sampling, not a conditional-infilling entry point): at each step, every still-masked position is stochastically resampled from the analytic posterior over the log-linear noise schedule via Gumbel-max categorical sampling, rather than chosen deterministically by confidence ranking. Conditioning/infilling needs no special-casing, since context tokens are simply never set to the mask token and the official update already leaves unmasked positions untouched. **Qwen2.5-Coder-0.5B** [9] is the autoregressive FIM baseline, chosen at the smallest available Qwen2.5-Coder scale ($\sim$500M parameters) to keep the two systems close in size and avoid confounding the AR-vs-diffusion comparison with a scale gap; under random token masking, each masked word is restored via its own independent single-hole FIM prompt (rather than one combined multi-hole prompt), since standard FIM prompting only supports a single gap per call. Each prompt's prefix/suffix context is built from the ground-truth (uncorrupted) document, and the N resulting single-hole prompts for a document are issued in fixed-size batches rather than sequentially; this choice is dictated by the computational constraints of a single Colab T4 GPU, where running N fully autoregressive, context-dependent calls one after another was prohibitively slow, not by a modeling decision favoring either system. For both systems, the context fed to the model around a mask is capped at 75 words on each side, bounding prompt/sequence length – and thus latency and memory – independently of document length, and keeping the two systems' effective context comparable. This yields an ablation grid of 100 documents $\times$ 2 corruption topologies $\times$ 3 masking ratios $\times$ (1 Qwen configuration + 3 MDLM timestep settings) $= 2\,400$ evaluated restorations in total. All real-model inference was run on a single Google Colab T4 GPU.

Measures and protocol. Each restored document is scored against the original, uncorrupted document with the three automatic metrics motivated in Section 5: ROUGE-L F1 [4] for surface-level n-gram/subsequence overlap, BERTScore F1 [10] (RoBERTa-large backbone) for embedding-based

Model	ROUGE-L	BERTScore	PPL	Lat. (s)
Qwen-0.5B	0.445	0.863	16.2	5.74
MDLM	0.752	0.938	143.3	0.31

Table 1: Mean ROUGE-L F1, BERTScore F1, geometric-mean perplexity (PPL), and per-restoration latency, averaged over the full ablation grid.

Model	Ratio	ROUGE-L	BERTScore
Qwen-0.5B	10%	0.580	0.885
	30%	0.425	0.860
	50%	0.331	0.843
MDLM	10%	0.923	0.979
	30%	0.759	0.939
	50%	0.574	0.895

Table 2: Mean ROUGE-L F1 and BERTScore F1 by masking ratio, averaged over both corruption topologies.

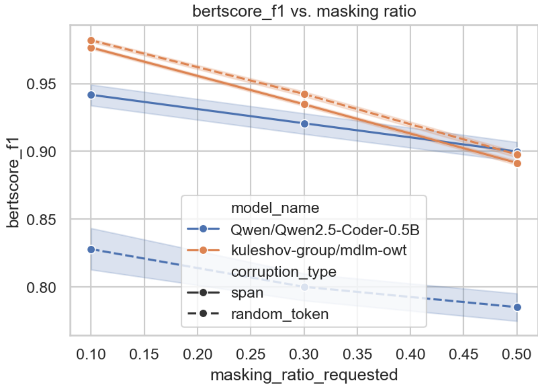

Figure 1: BERTScore F1 vs. masking ratio, by model and corruption topology (solid: span masking; dashed: random token masking).

semantic similarity, and perplexity of the reconstructed document under an external, independently trained GPT-2 model [7], mirroring the fluency measure used in the diffusion literature [8]. All three metrics are computed on the fully reinserted document (masked spans filled back into their original positions) rather than on the fill content in isolation, so that restorations are scored in the same bidirectional context they were generated for. Scores are aggregated by averaging across the grid's 100 documents within each (model, corruption type, masking ratio, [timesteps]) cell, which is the granularity used for the analysis in Section 6.

6.1 Results

Global comparison. Table 1 reports each system's scores averaged over the full ablation grid. MDLM obtains higher ROUGE-L and BERTScore than Qwen2.5-Coder-0.5B, and restores each document roughly 20× faster. Perplexity, reported here as a geometric mean (Section 6.2) to limit the influence of a small number of extreme-perplexity restorations, is higher for MDLM than for Qwen.

Effect of masking ratio. Table 2 breaks the two quality metrics down by masking ratio (averaged over both corruption topologies). Both systems' ROUGE-L and BERTScore decrease monotonically as the masking ratio increases from 10% to 50%. The gap between the two systems is instead largest at 10% (ROUGE-L: 0.580 vs. 0.923) and narrows monotonically as the masking ratio increases, to 0.331 vs. 0.574 at 50%.

Figure 1 shows this same ratio effect broken down further by corruption topology. For Qwen, the span-masking curve sits consistently above the random-token curve by a roughly constant margin (BERTScore: ∼0.11–0.12) at all three ratios. For MDLM, the two curves are close together at every ratio, with random-token masking scoring marginally higher than span masking throughout (BERTScore: 0.982 vs. 0.977 at 10%, 0.942 vs. 0.935 at 30%, 0.898 vs. 0.892 at 50%).

Effect of corruption topology. Figure 2 breaks ROUGE-L and BERTScore down by corruption topology. Qwen's scores drop sharply from span masking to random token masking (ROUGE-L: 0.672 → 0.219; BERTScore: 0.921 → 0.804), while MDLM's scores are comparable across the two topologies and are, on average, slightly higher under random token masking than under span masking (ROUGE-L: 0.740 → 0.765; BERTScore: 0.934 → 0.941).

Effect of denoising timesteps (MDLM). Figure 3 shows MDLM's BERTScore F1 and per-restoration latency as a function of the number of denoising timesteps (4, 8, 16), at each masking ratio. BERTScore F1 is nearly flat across this range (within 0.0010–0.0032 of its 16-timestep value across the three ratios, with the smallest deviation at 10% masking and the largest at 50%), while latency grows roughly linearly with the number of timesteps, from a mean of 0.139s at 4 steps to 0.52s at 16 steps.

6.2 Discussion

Why MDLM outperforms Qwen on overlap and semantic metrics. MDLM's advantage on ROUGE-L and BERTScore (Table 1) is consistent with the architectural difference outlined in Section 3: MDLM denoises every masked po-

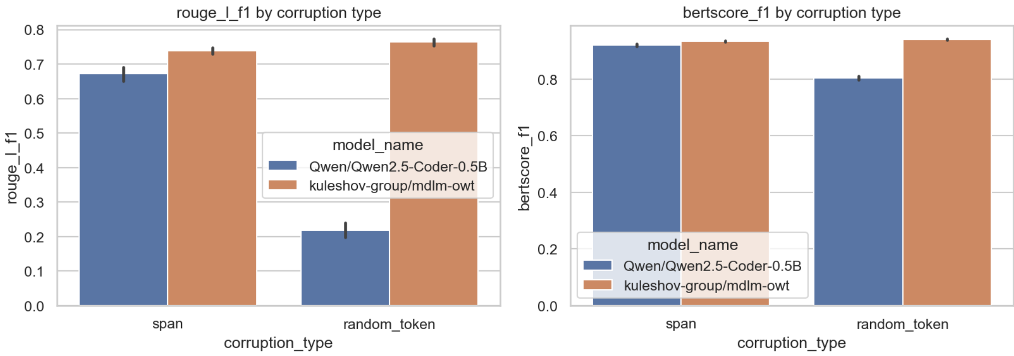

Figure 2: ROUGE-L F1 (left) and BERTScore F1 (right) by corruption topology, for both systems, averaged over all masking ratios.

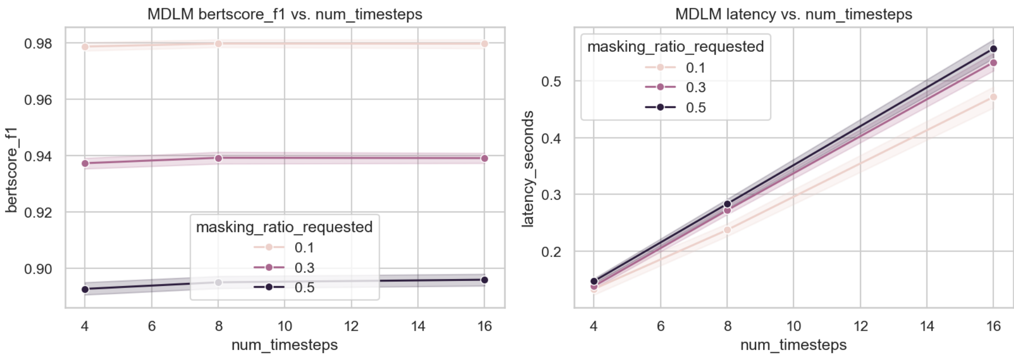

Figure 3: MDLM BERTScore F1 (left) and per-restoration latency (right) as a function of the number of denoising timesteps, by masking ratio.

sition jointly, in a single forward pass conditioned on the entire (fixed) bidirectional context, whereas Qwen's FIM prompting only ever conditions on a local prefix/suffix window and, for 'random_token' masking, resolves each masked word independently of every other mask in the same document (Section 6). MDLM's absorbing-state formulation also guarantees exact recovery of the original token count at each masked position, while Qwen's generated fill is free-length and can under- or over-shoot the true span, directly penalizing ROUGE-L's overlap-based scoring.

Why Qwen collapses under random token masking. The corruption-topology breakdown (Figure 2, Figure 1) shows this gap is not uniform: Qwen loses far more quality under random token masking than under span masking, while MDLM is largely insensitive to topology. This is a direct consequence of how each system handles multiple, scattered masks. Under 'random_token' masking, Qwen restores each masked word via its own single-hole FIM call built against the true original context around that one word (Section 6) – each call is individually easy, but the calls are independent, so nothing prevents two nearby restorations from being locally plausible yet jointly inconsistent (e.g., mismatched number/tense agreement across two masked words in the same clause), and errors from one call cannot be corrected using another. MDLM instead resolves all masked positions for a document in one joint reverse-diffusion pass, so the model's own distribution over one masked position is implicitly conditioned on its current belief about the others, giving it a structural advantage on exactly this corruption pattern.

Why MDLM's perplexity is higher despite better overlap/semantic scores. MDLM scores worse than Qwen on perplexity (Table 1) even though it scores better on ROUGE-L and BERTScore. A plausible explanation is that perplexity here is measured under an external GPT-2 model (Section 6) that is itself autoregressive: it may reward passages that resemble left-to-right GPT-2 generation, which could favor Qwen's output (itself autoregressive) over MDLM's regardless of how faithful either restoration actually is to the masked content. Both systems' per-document perplexity distributions are also heavily right-skewed, with a small number of restorations reaching into the thousands, so the arithmetic mean sits well above the median for both (Qwen: 94.6 vs. 16.9; MDLM: 319.2 vs. 112.1) – likely driven by a handful of restorations with a disfluent junction at a mask boundary. For this reason, Table 1 instead reports the geometric mean (Qwen: 16.2; MDLM: 143.3), the standard, outlier-robust aggregation for perplexity.

Why MDLM quality is largely insensitive to the number of denoising timesteps. Figure 3 shows BERTScore F1 barely moves between 4 and 16 timesteps, while latency scales roughly linearly with the timestep count. This is plausibly a consequence of our masking ratios (10–50%) leaving a majority of each document's tokens fixed as conditioning context: with that much unmasked scaffolding, the reverse-diffusion posterior over the remaining masked positions is already fairly peaked after only a few denoising steps, so additional steps mostly refine token choices at the margin rather than resolving substantial uncertainty. This suggests that, at least in this masking-ratio range, MDLM's fastest configuration (4 timesteps) is close to Pareto-optimal for this task, and the inference-efficiency axis may matter more at masking ratios beyond those tested here.

Qualitative error inspection. We inspect concrete restorations from a small illustrative run (20 WikiText-103 documents, 'random_token' masking, 30% ratio) to ground the patterns above. Documents in this run were capped at 20 words, rather than the 500-word cap used for the quantitative grid in Sections 6–6.1, so that full (corrupted, restored, original) triples stay short enough to quote inline in this section. This shorter cap gives both systems less context per mask than in the main experiment, so the examples below should be read as a lower bound on each system's quality, not as directly comparable to the aggregate numbers in Table 1. On one document, Qwen's fill for a document with several scattered masks derails into unrelated boilerplate rather than restoring the original words:

> **Original:** *"A " Summary of the Work Done for November , 1862 , Little Rock Arsenal " shows : Fabrication :"*
>
> **Qwen (worst case, BERTScore 0.73):** *"# -*- coding: utf-8 -*- [...] Created on Mon Nov 22 14:56:20 2019 [...] is a 1912 novel by American author William H. H. S. `` H. S. '' H. S. `` H. S. '' [...] <|fim_pad|> (repeated ∼60 times) [...] , 1862 , Little Rock Arsenal '' : Fabrication :"*

The model fabricates a Python file header, repeats fragments of the visible context verbatim, invents an unrelated "1912 novel" attribution, and pads the remainder with a special token rather than terminating – exactly the kind of jointly-inconsistent, uncoordinated errors predicted above for independent single-hole FIM calls, and a plausible symptom of Qwen2.5-Coder's own FIM training distribution being code, not natural language. Notably, this restoration still receives a BERTScore F1 of 0.73 (and even a garbled but shorter Qwen restoration on another document reaches BERTScore 0.89 while containing repeated, incoherent fragments), underscoring that embedding-based similarity is only a partial proxy for human-perceived restoration quality.

MDLM's best-scoring restoration in the same run, by contrast, recovers the original almost verbatim from six scattered masks:

> **Original:** *"In 1997 , Fey and other members of The Second City provided voices for the pinball game Medieval Madness ."*
>
> **MDLM (BERTScore 0.98):** *"In 1997 , he and other members of the Second City provided voices for the pinball game Medieval Madness ."*

The only deviations are a plausible pronoun in place of the proper noun "Fey" (unrecoverable from context alone, since nothing in the visible text names the referent) and a lower-cased "the" – both minor relative to Qwen's failure mode on the same corruption topology. MDLM's own worst case in this run is less dramatic than Qwen's: it substitutes plausible but incorrect content ("Cherry Cherry Makes You Happy" for "Wild Cherry Makes A Wish") while remaining a well-formed, on-topic sentence, rather than derailing into unrelated text.

Limitations. Several factors limit how far these results generalize. *Scale mismatch:* Qwen2.5-Coder-0.5B (∼494M parameters) is still roughly 4× larger than MDLM (∼125M); it is the closest available size match (Section 3) but not an exact one, so part of the residual gap in either direction may reflect capacity rather than paradigm. *Domain mismatch:* 'kuleshov-group/mdlm-owt' is trained on OpenWebText, not on Wikipedia-style text, whereas Qwen2.5-Coder is trained (and FIM-tuned) primarily on code; neither system was fine-tuned on WikiText-103 or on restoration, so both are evaluated zero-shot but under different degrees of domain shift from the evaluation corpus. *Residual-mask cleanup at low timestep counts:* our reverse process ports the official `kuleshov-group/mdlm` reverse-diffusion update (stochastic Gumbel-max sampling from the analytic posterior over the log-linear noise schedule) rather than approximating it with a deterministic heuristic, so the reported MDLM results reflect the model's genuine sampler; the one remaining deviation from [8] is that any position still masked after our few-timestep reverse loop (where the schedule's residual stay-masked probability is non-negligible) is finalized with one extra forward pass and an argmax over non-mask logits – a practical cleanup step (`src/mdlm_utils.py`), not part of the ported sampling algorithm. *External, asymmetric perplexity metric:* as discussed above, the GPT-2 perplexity metric structurally advantages Qwen's autoregressive output, so cross-paradigm perplexity comparisons should be read with that asymmetry in mind. *Sample size and human evaluation:* results are averaged over 100 WikiText-103 documents per grid cell, with no human judgment of fluency or adequacy collected, so the automatic-metric gaps reported here (Section 6.1) have not been validated against human preference. *Single-sample decoding:* both systems generate one restoration per corrupted document (Qwen with temperature 0.2, no beam search or best-of-n; MDLM with a single reverse-diffusion trajectory), so the reported scores reflect one draw from each system's output distribution rather than its best-achievable quality. *Metric blind spots:* the qualitative examples above show BERTScore F1 remaining as high as 0.73–0.89 for Qwen restorations containing repeated fragments, fabricated content, or non-terminating pad tokens, indicating the automatic metrics used here do not fully penalize this class of gross, structurally obvious failure. *Optimistic per-call context for Qwen:* under random token masking, each of Qwen's independent FIM calls sees the true, uncorrupted word at any other masked position within its window rather than a placeholder – an idealization that in isolation should only inflate Qwen's per-call accuracy relative to MDLM. Yet Qwen's scores still collapse under this topology (Section 6.2), so the lack of coordination across independent calls evidently outweighs this advantage; the true gap attributable to the AR-vs-diffusion coordination difference alone is therefore, if anything, larger than the reported numbers suggest.

7 Conclusions

This work compared zero-shot text restoration under two paradigms – masked diffusion (MDLM) and autoregressive Fill-in-the-Middle prompting (Qwen2.5-Coder-0.5B) – across a controlled ablation grid of masking ratio, corruption topology, and (for MDLM) denoising timesteps, on synthetically corrupted WikiText-103 documents (Sections 6–6.2). MDLM outperformed the size-matched AR baseline on both ROUGE-L and BERTScore at every masking ratio and under both corruption topologies, while being roughly 20× faster per restoration (Table 1); its advantage was largest under random token masking, where Qwen's independent single-hole FIM calls produced jointly inconsistent fills, while MDLM's single joint reverse-diffusion pass resolved all masked positions coherently (Figure 2). The only metric favoring Qwen, perplexity, is confounded by its own autoregressive scoring model and should not be read as a paradigm-neutral fluency measure (Section 6.2). Together, these results support the hypothesis motivating this project (Section 3): the architectural property that lets masked diffusion enforce hard bidirectional constraints natively, rather than approximate them through prompting, translates into a measurable empirical advantage for zero-shot text restoration, at least at the model scales and masking ratios (10–50%) tested here. At the same time, MDLM's quality was found to be nearly insensitive to the number of denoising timesteps in this range (Figure 3), a finding an extended sweep confirms holds from single-step decoding up to 128 steps under random token masking at 30% masking ratio (Appendix A).

Future work. Several directions follow directly from the limitations identified in Section 6.2. *MDLM timestep breakdown point:* Appendix A extends the timestep sweep to $\{1, 2, 16, 32, 64, 128\}$ at a fixed masking ratio (30%, random token masking) and finds quality remains essentially flat throughout, so the breakdown point hypothesized here was not observed even across two orders of magnitude in timesteps; a natural next step is to test whether it instead emerges along the masking-ratio axis (e.g., up to 90%), where a diffusion model has much less bidirectional context left to condition on, or with an even smaller, more parameter-constrained MDLM checkpoint. *Scale-matched comparison:* repeating this ablation with AR and diffusion checkpoints trained on more closely matched data and parameter counts (rather than the closest publicly available sizes used here) would isolate the paradigm effect from the residual scale and domain confounds noted in Section 6.2. *Metric and human evaluation:* given that BERTScore assigned scores as high as 0.73–0.89 to visibly degenerate Qwen restorations (Section 6.2), future work should incorporate human judgments of fluency and adequacy, and/or metrics more sensitive to repetition and non-termination, alongside the automatic metrics used here. *Broader corruption and domain settings:* extending the ablation to additional corruption topologies (e.g., multiple disjoint spans), non-Wikipedia domains, and multiple sampled restorations per document (rather than single-sample decoding) would test how far the trends reported here generalize beyond the present experimental setup. *Fairer random-token comparison for Qwen:* the batched, ground-truth-conditioned FIM calls used here for Qwen were adopted for computational reasons and give it an idealized per-call context under random token masking (Section 6.2); a follow-up could instead issue Qwen's calls sequentially, conditioning each on the document as restored so far rather than on the ground truth, at the cost of the speed gained from batching – isolating the AR-vs-diffusion coordination gap from this confound.

Use of GenAI and LLM tools. Claude Code (Anthropic) was used throughout this project as a coding assistant: implementing and refactoring the modules in `src/` (data corruption, model wrappers, evaluation pipeline) under the author's direction and review, drafting/editing portions of this report's prose based on the author's outline and experimental results, and assisting with debugging. Claude and Gemini (Google) were additionally used earlier in the project to research and study the task, i.e. to explore the masked-diffusion and Fill-in-the-Middle literature and clarify candidate experimental designs before they were implemented. All experimental design choices, the code's correctness, and the content of this report remain the author's responsibility; no citation, number, or claim in this report was generated without being checked by the author against the code or the underlying literature.

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A Extended timestep ablation for MDLM

Setup. Section 6.2 found MDLM's quality nearly insensitive to the number of denoising timesteps over the main grid's 4–16 range. To test whether this holds well beyond that range, this appendix reports a dedicated, MDLM-only sweep at random token masking, 30% masking ratio – the same corruption type and ratio used in the qualitative error inspection above (Section 6.2) – reusing the same 100 WikiText-103 documents (fixed seed 42) and 75-word context window as the main grid (Section 6), extended to `num_timesteps` $\in \{1, 2, 16, 32, 64, 128\}$, yielding 600 restorations.

Results. Table 3 and Figure 4 show BERTScore F1 (left) and per-restoration latency (right) across this range. BERTScore F1 stays within a narrow 0.940–0.945 band across the full two-orders-of-magnitude sweep (0.940 at 1 step vs. 0.943 at 128 steps, peaking at 0.945 at 64 steps), with no monotonic trend distinguishable from run-to-run variation; ROUGE-L F1 is similarly flat (0.773–0.777, a range of 0.004). Latency instead grows linearly with timesteps, from 0.06s at 1 step to 4.25s at 128 steps – a 66$\times$ increase in compute for a swing in BERTScore F1 that stays within noise. Perplexity is the one metric with a discernible trend: its geometric mean (the same outlier-robust aggregation used in Table 1) falls from 169.5 at 1 step to 118.2 by 16 steps, then stays essentially flat through 128 steps (114.8) – the only sign of an accuracy/efficiency trade-off in this sweep, and it saturates well within the main grid's own tested range. The shared 16-timestep setting closely reproduces this same cell from the main grid (BERTScore 0.943 here vs. the 0.942 reported for `random_token` masking at 30% in Figure 1), a useful consistency check between the two independently sampled runs.

Timesteps	ROUGE-L	BERTScore	PPL
1	0.777	0.940	169.5
2	0.773	0.942	132.1
16	0.776	0.943	118.2
32	0.774	0.943	116.5
64	0.777	0.945	117.1
128	0.777	0.943	114.8

Table 3: MDLM ROUGE-L F1, BERTScore F1, and geometric-mean perplexity (PPL) by number of denoising timesteps, random token masking, 30% masking ratio.

Discussion. This extends the finding from Section 6.2 across two orders of magnitude in denoising steps, from single-step decoding up to 128 steps, at random token masking, 30% masking ratio. Quality remains essentially flat throughout: even one-step decoding is already close to whatever ceiling additional steps can reach, and the accuracy/efficiency trade-off motivating the timesteps ablation axis does not manifest at this masking ratio and corruption type even 8$\times$ beyond the main grid's highest setting. This reinforces few-step decoding as a robust, near-Pareto-optimal default for MDLM-based restoration under this checkpoint and corruption topology, though it remains untested whether this insensitivity persists at masking ratios above 50%, where the model has much less bidirectional context left to condition on.

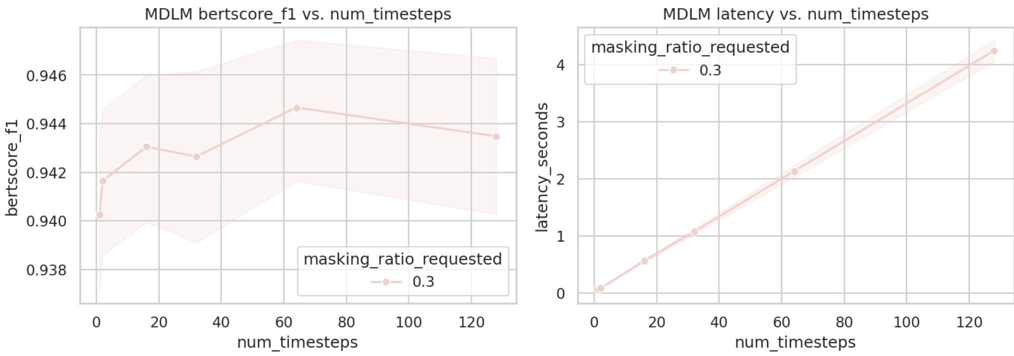

Figure 4: MDLM BERTScore F1 (left) and per-restoration latency (right) as a function of the number of denoising timesteps (1–128), random token masking, 30% masking ratio.